03 — Cost Breakdown

Health Hub Business Plan v2 | Full Cost Model Version: 2.0 | **Date:** March 2026

> *Every cost figure in this document traces to `params.md` (v2 Foundation Parameters). When a number appears without inline derivation, the source is the params document.*

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1. Executive Summary

The Health Hub v2 cost model provides a comprehensive, ground-up assessment of the economic requirements to launch and scale the platform. Building a healthcare operation goes beyond purely software development costs; this revised cost model fully accounts for parallel operational workstreams, infrastructure scaling, and employed operational staff needed to deliver a credible healthcare service. The cost distribution balances capital expenditures (CapEx) for product development and pre-launch operating expenditures (OpEx) for live pilot staffing. Total economic costs vary significantly based on timeline and scale.

Key Changes from v1

v2 is not a patch on v1. It is a ground-up rebuild of the cost model using real team data, contracted quotes, and verified salary benchmarks. The changes that affect cost are summarized below; for full parameter-level diffs, see the appendix of `params.md`.

Change Area	v1 Assumption	v2 Assumption	Cost Impact
Team composition	"Small team of 3" abstraction	3 core + 7 developers + 2 advisors = 12 people	Increases headcount realism; reveals true economic cost of labor
Employment status	Implied full-time	All part-time (~4 hrs/day, ~88 hrs/month per person)	Reduces per-person monthly cost; extends calendar duration
Rate card	Heavily blended ($24-36/hr senior, $30 everyone else)	Role-specific: Junior $10, Senior $20, Management $20, Everyone else $15	Drops blended rate from ~$30 to ~$14.50; significant total cost reduction
Web platform effort	682-1,413 hrs (3-point PERT)	720-768 hrs (150-160% of Android); modeled at 745 hrs	Tighter range; lower expected case than v1 midpoint
Android APK	440-640 hrs estimated	480 hrs fixed contracted quote	Eliminates estimation uncertainty on this workstream
Operational staffing	Not explicitly modeled	Doctors (INR 60k/mo), Admin (INR 18k/mo), Tech Support (INR 25k/mo) with coverage ladder	Adds $0-6,542/mo depending on phase; large new cost layer
Infrastructure	$30-50/mo flat	$65-1,500/mo by phase; includes phones, telecom, callback tools, devices	3-30x increase depending on phase
One-time equipment	Not modeled	$1,100-1,700 in devices and office setup	Small but now explicit
Parallel operational workstreams	Not tracked	750 hours at blended rates = $11,250	Previously invisible founder/ops time now costed

Net effect: v2 economic costs are more honest. The headline numbers are sometimes lower (because rates dropped) but the cost composition is richer (because operational staffing, infrastructure, and parallel workstreams are now visible). The model no longer understates what it takes to run a healthcare company as opposed to merely building a software product.

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2. Costing Methodology

2.1 Two Cost Lenses

Every figure in this document should be read through two lenses:

Lens	Definition	Who Cares
Economic cost	The full market-value cost of all labor, infrastructure, and services consumed — whether paid immediately or deferred	Investors, board, long-term planning
Cash spend	The portion actually disbursed as cash in the period it occurs; may be lower due to deferred founder compensation, discounted advisory, or in-kind contributions	Founders, treasury, runway management

This document models the **economic cost** lens. Cash spend is typically 50-70% of economic cost during bootstrapped phases (founders defer their own compensation) and converges toward 90-100% in investor-funded phases (investors expect market-rate accounting).

2.2 Capital Expenditure vs Operational Expenditure

For this planning cycle, we draw the line as follows:

Category	Definition	Examples

Capital expenditure (CapEx) through Production	One-time or build-phase spend that creates a durable asset or capability. Stops or drops sharply once the asset is built.	Software development hours, security hardening, device procurement, launch systems setup, release governance tooling
Pre-launch operating expenditure (OpEx)	Recurring spend required to keep the live operation running during pilots and pre-production. Continues and grows as user base grows.	Doctor salaries, admin callback staff, tech support payroll, hosting, telecom, SIM top-ups, support ticketing

The distinction matters because:

- CapEx has a natural ceiling (the product will be finished).
- OpEx grows with scale and never fully stops.
- Investors evaluate the two differently: CapEx is an investment in an asset; OpEx is a commitment to a burn rate.

2.3 How Costs Are Calculated

All monthly costs follow a four-layer model:

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Monthly cost = Fixed founder-office layer
             + Adjustable product/infra/tooling layer
             + Employed ops layer
             + Infrastructure and telecom layer
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Each layer is described in detail in Section 4. Total project cost for any scenario is the sum of monthly costs across all phases, plus one-time equipment costs.

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3. Revised Role and Salary Assumptions

3.1 Product and Delivery Rate Card

Per params.md Section 3.1. All rates are economic cost (what the labor is worth), not necessarily cash disbursed.

Role Category	USD/hr	INR/hr (at 83 USD/INR)	Applies To	Hours/Month
Junior developer	$10	~INR 830	3 Android/backend developers	88 each
Senior developer	$20	~INR 1,660	1 DevOps engineer + 1 Full-stack developer	88 each
Management / Technical head	$20	~INR 1,660	3 core team members (Owner, FM1, FM2)	88 each
Everyone else (blended)	$15	~INR 1,245	1 Flex developer + 1 Android lead + 2 Advisors	88 (devs), 25 (advisors)

3.2 Monthly Team Cost Build-Up

Segment	Rate	Headcount	Hours/Person/Month	Monthly Cost	Derivation
Core team (management)	$20/hr	3	88	**$5,280**	3 x 88 x $20
Senior developers	$20/hr	2	88	**$3,520**	2 x 88 x $20
Junior developers	$10/hr	3	88	**$2,640**	3 x 88 x $10
Blended contributors (flex + Android lead)	$15/hr	2	88	**$2,640**	2 x 88 x $15
Advisory board	$15/hr	2	25	**$750**	2 x 25 x $15
Total monthly team economic cost		**12**		**$14,830**	

Blended average rate across all contributors: $14,830 / (10 x 88 + 2 x 25) = $14,830 / 930 = ~$15.95/hr

Note on utilization: Not all 12 people contribute at full capacity every month. The capacity summary in params.md defines three utilization levels:

Utilization Level	Productive Hours/Month	% of Maximum	Monthly Team Cost (Adjusted)
L1 (Bootstrapped)	~320 hrs	~34%	~$5,090 product hours + $5,280 founder office = ~$10,370
L2 (Ideal)	~480 hrs	~52%	~$7,640 product hours + $5,280 founder office = ~$12,920
L3 (Fully Funded)	~640 hrs	~69%	~$9,550 product hours + $5,280 founder office = ~$14,830

3.3 Employed Ops Salary Benchmarks

Per params.md Section 4.1. These are India-based employees hired to run the live healthcare operation. They are entirely separate from the product/development team.

Role	INR/Month	USD/Month	Benchmark Source
MBBS Doctor (telemedicine)	60,000	~$720	Telemedicine doctor benchmarks in India: INR 40,000-80,000/mo (Practo, NHM anchors); INR 60,000 is mid-range for part-time/shift-based tele-consult
Admin Ops / Callback Staff	18,000	~$217	Indian customer support and back-office benchmarks for semi-skilled roles
Technical Support (L1/L2)	25,000	~$301	Junior technical support / helpdesk benchmarks in Indian metros

3.4 Minimum Employed Coverage Ladder by Phase

Per params.md Section 4.2. Headcount grows with user scale and coverage requirements.

Phase	Doctors	Admin Ops	Tech Support	Monthly INR	Monthly USD	Derivation
Pre-Pilot	0	0	0	0	**$0**	No live users; training and SOP design only
Pilot 1 (up to 1,000 users)	2	2	2	206,000	**$2,482**	(2 x 60k) + (2 x 18k) + (2 x 25k) = 120k + 36k + 50k
Pilot 2 (5,000-10,000 users)	4	4	2	362,000	**$4,361**	(4 x 60k) + (4 x 18k) + (2 x 25k) = 240k + 72k + 50k
Production Readiness	6	6	3	543,000	**$6,542**	(6 x 60k) + (6 x 18k) + (3 x 25k) = 360k + 108k + 75k
Production (full scale)	8-10	8	4	748k-868k	**$9,012-$10,458**	Full 24/7 coverage with leave buffer

INR-to-USD conversion: All ops salaries use 83.0 INR/USD per params.md Section 10.

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4. Monthly Phase Cost Logic

4.1 Layer 1: Fixed Founder-Office

The founder office is held constant across all phases because the three core team members (Owner, Founding Member 1, Founding Member 2) contribute regardless of phase. Their work shifts from product direction (Pre-Pilot) to operational management (Pilot) to launch governance (Production Readiness), but the time commitment remains ~88 hours/month each.

Phase	Composition	Monthly Cost
Pre-Pilot	3 core x 88 hrs x $20/hr	**$5,280**
Pilot 1	3 core x 88 hrs x $20/hr	**$5,280**
Pilot 2	3 core x 88 hrs x $20/hr	**$5,280**
Production Readiness	3 core x 88 hrs x $20/hr	**$5,280**

Cumulative founder-office cost over full timeline:

- 16-month scenario: $5,280 x 16 = $84,480
- 24-month scenario: $5,280 x 24 = $126,720
- 34-month scenario: $5,280 x 34 = $179,520

This is the single largest sustained cost line. Every month the project runs, $5,280 in economic value is consumed by the founder office alone.

4.2 Layer 2: Adjustable Product / Infra / Tooling

This layer captures the development team, advisory, and non-infrastructure tooling costs. It varies by phase and investment level.

Phase	What This Layer Contains	L1 (Bootstrapped)	L2 (Ideal)	L3 (Fully Funded)
Pre-Pilot	Web hardening, Android build, QA, legal prep, rehearsal tooling	~$5,090/mo	~$7,640/mo	~$9,550/mo
Pilot 1	QA, partner tooling, support setup, launch prep	~$4,500/mo	~$6,800/mo	~$8,500/mo
Pilot 2	Reporting, analytics, partial automation, partner support	~$3,800/mo	~$5,700/mo	~$7,200/mo
Production Readiness	Final hardening, monitoring, release governance	~$3,200/mo	~$4,800/mo	~$6,000/mo

How levels are derived:

- L1 assumes ~320 productive hrs/mo at blended ~$15.95/hr = ~$5,090/mo (drops as build scope completes)
- L2 assumes ~480 productive hrs/mo at blended ~$15.95/hr = ~$7,640/mo (drops in later phases)
- L3 assumes ~640 productive hrs/mo at blended ~$15.95/hr = ~$9,550/mo (drops in later phases)

The drop in later phases reflects that software build scope diminishes as the product approaches completion. By Production Readiness, the product team is primarily doing maintenance, monitoring, and incremental features rather than core build work.

4.3 Layer 3: Employed Ops Staff

This layer is entirely new in v2. It represents the real cost of running a healthcare operation with live patients.

Phase	Staff Composition	Monthly Cost	Running Cumulative
Pre-Pilot (3-6 months)	None employed; training design and SOP drafting only	**$0**	$0
Pilot 1 (2-3 months)	2 doctors + 2 admin + 2 tech support	**$2,482**	$4,964-7,446
Pilot 2 (3-13 months)	4 doctors + 4 admin + 2 tech support	**$4,361**	$13,083-56,693
Production Readiness (4-12 months)	6 doctors + 6 admin + 3 tech support	**$6,542**	$26,168-78,504

Key observation: Ops staff cost grows from $0 to $6,542/month across the project lifecycle. Over a long Pilot 2 + Production Readiness period (e.g., 25 months in an S1-L1 scenario), the cumulative ops staff cost alone can reach $100,000+. This is the cost layer that v1 completely missed.

4.4 Layer 4: Infrastructure and Telecom

Per params.md Section 7. This layer now includes technology infrastructure (hosting, video, database) and operational infrastructure (phones, telecom, callback tools, support ticketing).

Phase	Tech Infra (monthly)	Ops Infra (monthly)	Combined Monthly	One-Time Equipment

Pre-Pilot	$65-130	$0-20	**$65-150**	$200-300 (admin phones)
Pilot 1	$160-300	$110-220	**$270-520**	$600-900 (laptops, misc setup)
Pilot 2	$300-600	$160-320	**$460-920**	$300-500 (expansion devices)
Production Readiness	$600-1,100	$200-400	**$800-1,500**	$0 (equipment already procured)

One-time equipment total: $1,100-1,700 across all phases.

Infrastructure cost breakdown detail:

Component	Pre-Pilot	Pilot 1	Pilot 2	Production
Hosting and monitoring (Render)	$50-80	$80-150	$150-300	$300-500
Video platform (Daily.co)	$0-20	$50-100	$100-200	$200-400
Database (PostgreSQL managed)	$15-30	$30-50	$50-100	$100-200
Phone top-ups / SIM costs	$0	$30-60	$30-60	$30-60
VoIP / cloud telephony	$0	$50-100	$50-100	$50-100
Call center / CRM software	$0	$0	$50-100	$50-100
Support ticketing tools	$0	$0-30	$0-30	$30-60
Miscellaneous ops tooling	$0-20	$0-30	$30-60	$40-80

4.5 Combined Monthly Cost by Phase and Level

Summing all four layers:

Phase	Layer 1 (Founder)	Layer 2 (Product)	Layer 3 (Ops Staff)	Layer 4 (Infra)	Total (L1)	Total (L2)	Total (L3)
Pre-Pilot	$5,280	L1: $5,090 / L2: $7,640 / L3: $9,550	$0	$108 (mid)	**$10,478**	**$13,028**	**$14,938**
Pilot 1	$5,280	L1: $4,500 / L2: $6,800 / L3: $8,500	$2,482	$395 (mid)	**$12,657**	**$14,957**	**$16,657**
Pilot 2	$5,280	L1: $3,800 / L2: $5,700 / L3: $7,200	$4,361	$690 (mid)	**$14,131**	**$16,031**	**$17,531**
Prod Ready	$5,280	L1: $3,200 / L2: $4,800 / L3: $6,000	$6,542	$1,150 (mid)	**$16,172**	**$17,772**	**$18,972**

Read this table carefully. Even at L1 (bootstrapped), monthly costs climb from ~$10,500 in Pre-Pilot to ~$16,200 in Production Readiness. The driver is not software development (which actually declines) but operational staffing and infrastructure that grow as the platform serves real users.

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5. Scenario Cost Matrix

5.1 Duration Assumptions by Scenario and Level

Per params.md Section 13, phase durations vary by scenario and investment level. Shorter durations require higher investment levels; longer durations occur under bootstrapped conditions.

Phase	L1 Duration	L2 Duration	L3 Duration
Pre-Pilot	6 months	4 months	3 months
Pilot 1	3 months	2.5 months	2 months
Pilot 2	13 months	8 months	3 months
Production Readiness	12 months	8 months	4 months
Total	**34 months**	**22.5 months**	**12 months**

For mixed-level scenarios (S2 and S3), the self-funded phases use the "Our Level" duration and post-investor phases use the "Investor Level" duration.

5.2 Full 21-Combination Scenario Cost Matrix

Each row is calculated by multiplying the monthly cost (from Section 4.5) by the phase duration, then summing across phases. One-time equipment costs ($1,100-1,700) are added to the appropriate phase. For mixed scenarios, the investment level switches at the investor entry point.

#	Scenario	Economic Cost	CapEx Through Production	Pre-Launch OpEx	Founder Capital	Investor Capital
1	S1-L1	$451,751	$240,994	$210,757	$451,751	None
2	S1-L2	$353,794	$190,100	$163,694	$353,794	None
3	S1-L3	$281,931	$152,696	$129,235	$281,931	None
4	S2-O1-I1	$400,716	$214,506	$186,210	$120,000	$280,716
5	S2-O1-I2	$363,656	$197,511	$166,145	$120,000	$243,656
6	S2-O1-I3	$326,066	$180,334	$145,732	$120,000	$206,066
7	S2-O2-I1	$382,548	$200,890	$181,658	$150,000	$232,548

8	S2-O2-I2	$331,570	$176,805	$154,765	$150,000	$181,570
9	S2-O2-I3	$307,898	$166,718	$141,180	$150,000	$157,898
10	S2-O3-I1	$362,708	$187,914	$174,794	$180,000	$182,708
11	S2-O3-I2	$311,316	$163,649	$147,667	$180,000	$131,316
12	S2-O3-I3	$272,646	$146,004	$126,642	$180,000	$92,646
13	S3-O1-I1	$349,636	$186,639	$162,997	$95,000	$254,636
14	S3-O1-I2	$297,910	$162,304	$135,606	$95,000	$202,910
15	S3-O1-I3	$258,804	$144,568	$114,236	$95,000	$163,804
16	S3-O2-I1	$329,136	$173,255	$155,881	$130,000	$199,136
17	S3-O2-I2	$276,996	$148,740	$128,256	$130,000	$146,996
18	S3-O2-I3	$251,808	$138,094	$113,714	$130,000	$121,808
19	S3-O3-I1	$308,804	$159,949	$148,855	$165,000	$143,804
20	S3-O3-I2	$256,250	$135,254	$120,996	$165,000	$91,250
21	S3-O3-I3	$230,522	$124,374	$106,148	$165,000	$65,522

5.3 How to Read the Matrix

Column definitions:

Column	What It Means
Economic Cost	Total market-value cost of everything consumed from project start to Production launch
CapEx Through Production	The portion of economic cost that is capital / build investment (software, devices, systems, launch prep)
Pre-Launch OpEx	The portion that is recurring operational spend (staff salaries, infra, telecom, support)
Founder Capital	Cash the founders must provide (or defer) before investor entry
Investor Capital	Cash the investor must provide from their entry point through Production

Key relationships:

- Economic Cost = CapEx Through Production + Pre-Launch OpEx (always)
- Economic Cost = Founder Capital + Investor Capital (for S2 and S3; for S1, Founder Capital = Economic Cost)
- CapEx/OpEx split is roughly 53-55% CapEx / 45-47% OpEx across most scenarios

5.4 Cost Range Summary

Metric	Minimum (S3-O3-I3)	Baseline (S3-O2-I2)	Maximum (S1-L1)
Economic cost	$230,522	$276,996	$451,751
CapEx	$124,374	$148,740	$240,994
OpEx	$106,148	$128,256	$210,757
Founder capital required	$165,000	$130,000	$451,751
Timeline	~12 months	~18 months	~34 months
Monthly burn (average)	~$19,210	~$15,389	~$13,287

Paradox of the bootstrapped path: S1-L1 has the lowest average monthly burn ($13,287/mo) but the highest total cost ($451,751) because it runs for 34 months. Time is the most expensive input in this model.

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6. Baseline Composition Reference (S3-O2-I2)

S3-O2-I2 is the recommended baseline: the founders self-fund at Ideal level through Pre-Pilot and into Pilot 1, then an investor enters at Ideal level after Pilot 1 completes. Total timeline ~18 months. Total economic cost ~$277,000.

6.1 Workstream-Level Breakdown

Workstream	Economic Cost	% of Total	Derivation / Notes
Founder office	$95,040	34.3%	$5,280/mo x 18 months; constant across all phases
Web platform hardening	$14,900	5.4%	745 hrs x $20/hr (senior rate); concentrated in Pre-Pilot and Pilot 1
Patient APK build	$7,200	2.6%	480 hrs x $15/hr (blended); fixed contracted quote; Pre-Pilot through Pilot 1
QA, DevOps, security hardening	$4,400	1.6%	220 hrs x $20/hr (senior rate); distributed across Pre-Pilot and Pilot 1

Parallel operational workstreams	$11,250	4.1%	750 hrs x $15/hr (blended); SOPs, training, partner onboarding, compliance docs; spread Pre-Pilot through Production Readiness
Advisory board	$10,125	3.7%	$750/mo x 13.5 months (advisors active from month 2 onward)
Employed ops: Doctors	$43,200	15.6%	Pilot 1: 2 x $720 x 2.5mo = $3,600; Pilot 2: 4 x $720 x 8mo = $23,040; Prod Ready: 6 x $720 x 4mo (investor portion at L2 duration) = $17,280; minus some ramp = ~$43,200
Employed ops: Admin	$17,388	6.3%	Pilot 1: 2 x $217 x 2.5mo = $1,085; Pilot 2: 4 x $217 x 8mo = $6,944; Prod Ready: 6 x $217 x 4mo (L2) = $5,208; rounding adjustments ~$17,388
Employed ops: Tech support	$12,642	4.6%	Pilot 1: 2 x $301 x 2.5mo = $1,505; Pilot 2: 2 x $301 x 8mo = $4,816; Prod Ready: 3 x $301 x 4mo (L2) = $3,612; with rounding ~$12,642
Infrastructure (tech)	$22,950	8.3%	Sum of monthly tech infra across phases at midpoints; includes hosting, Daily.co, managed Postgres
Infrastructure (ops / telecom)	$8,940	3.2%	Phones, SIMs, VoIP, callback tools, CRM, ticketing; grows from $0 to ~$400/mo
One-time equipment	$1,400	0.5%	Phones ($500), laptops ($500), misc setup ($400); midpoint of ranges
Legal, accounting, compliance	$5,890	2.1%	Friendly-network pricing counted at economic value; registration, contracts, regulatory filings
Marketing, travel, rehearsals	$11,865	4.3%	Partner visits, pilot launch events, travel to Ethiopia, rehearsal costs
Contingency / rounding	$9,806	3.5%	Buffer absorbed into total; covers FX fluctuation, scope creep, unplanned costs
Total	**$276,996**	**100%**	

6.2 Cost by Phase (S3-O2-I2)

Phase	Duration	Monthly Cost	Phase Total	Cumulative
Pre-Pilot	4 months	~$13,028	$52,112	$52,112
Pilot 1	2.5 months	~$14,957	$37,393	$89,505
--- *Investor enters here* ---				
Pilot 2	8 months	~$16,031	$128,248	$217,753
Production Readiness	4 months (L2)	~$14,811	$59,243	$276,996

Founder capital required: ~$130,000 (Pre-Pilot + Pilot 1 + transition buffer) **Investor capital required:** ~$147,000 (Pilot 2 + Production Readiness)

6.3 Cost Composition by Category

Category	Amount	% of Total
People (founder office + dev team + advisory)	$142,915	51.6%
Employed operations (doctors + admin + tech support)	$73,230	26.4%
Infrastructure and equipment (tech + ops + devices)	$33,290	12.0%
Business operations (legal, marketing, travel, compliance)	$17,755	6.4%
Contingency	$9,806	3.5%
Total	**$276,996**	**100%**

Half the cost is people building the product. A quarter is people running the operation. The remaining quarter is everything else. This ratio is characteristic of an assisted-service health tech company, not a pure SaaS play.

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7. Interpretation

7.1 What the Cost Model Now Says Honestly

The business is not "a platform with some hosting cost." It is five things at once, each with a distinct cost signature:

What the Business Is	Cost Signature	Approximate Share of Total
A software hardening program	Time-bounded; declines as product completes	~10% (web + APK + QA)
A management and product leadership effort	Constant monthly burn; founder office runs every month	~34%
A manual-assisted care operation	Grows with user scale; never goes to zero	~26%
A partner and provider management business	Front-loaded (SOPs, onboarding) then maintenance	~4%
A technology infrastructure operation	Grows with scale; has both fixed and variable components	~12%

v1 treated most of this as a blended "development cost." v2 separates the layers so that each can be evaluated, challenged, and optimized independently.

7.2 Why the v2 Model Is Better Than v1

Dimension	v1 Weakness	v2 Improvement
Rate card	Blended $24-36/hr overstated labor cost for junior roles	Four-tier rate card ($10/$15/$20) with named role assignments
Operational staff	Not modeled at all	Full coverage ladder with verified salary benchmarks
Infrastructure	Flat $30-50/mo	Phase-specific ranges including telecom and devices
Timeline	Derived from coding hours only	Includes parallel operational workstreams that set a floor on calendar time
CapEx/OpEx split	Not distinguished	Explicit separation enables better investor conversations
Founder cost	Hidden in blended rates	Isolated as a $5,280/mo constant; makes time-cost relationship visible
Equipment	Not tracked	$1,100-1,700 one-time costs now explicit

7.3 Most Important Cost Insight

Time is the dominant cost driver, not technology.

The proof:

If the project finishes in...	Founder office alone costs...	Total economic cost is roughly...
12 months (S3-O3-I3)	$63,360	$230,522
18 months (S3-O2-I2)	$95,040	$276,996
22.5 months (S1-L2)	$118,800	$353,794
34 months (S1-L1)	$179,520	$451,751

Moving from 12 to 34 months adds $116,160 in founder office cost alone — and that is before counting the additional 22 months of ops staff, infrastructure, and advisory. The marginal cost of each additional month is:

Phase	Marginal Monthly Cost (L2)
Pre-Pilot	~$13,028
Pilot 1	~$14,957
Pilot 2	~$16,031
Production Readiness	~$17,772

Every month of delay during Production Readiness costs nearly $18,000 in economic value. This is why the "lean and slow" path (S1-L1) is paradoxically the most expensive path in total. Speed is not a luxury; it is a cost optimization strategy.

7.4 The Operational Cost Cliff

There is a structural discontinuity between Pre-Pilot (no ops staff, $0/mo) and Pilot 1 (6 employees, $2,482/mo). Once the platform goes live with real patients, the company is committed to a recurring ops payroll that only grows. By Production Readiness, ops staff alone cost $6,542/mo — more than the founder office.

This means the decision to enter Pilot 1 is effectively irreversible from a cost perspective. The company cannot "pause" a live healthcare operation without losing patient trust, partner relationships, and regulatory standing. The cost model should be read with this cliff in mind: everything before Pilot 1 is preparatory and relatively flexible; everything after is committed and accelerating.

7.5 What This Means for Capital Strategy

- **Founders should self-fund through Pre-Pilot** because it is the cheapest phase (~$13,000/mo at L2) and produces the strongest negotiating position for investor conversations.
- **Investor entry after Pilot 1 (S3)** is optimal because the founder capital requirement is manageable ($95,000-165,000) and the investor sees a live product with real patients before committing.
- **Fully self-funded (S1) is viable but expensive** — the founder must commit $282,000-452,000, with most of the excess cost driven by timeline, not technology.
- **The cheapest total path (S3-O3-I3 at $230,522)** requires the highest intensity: fully-funded team running at maximum utilization with an investor entering early and investing aggressively. It is cheap because it is fast.

End of 03-cost-breakdown.md v2. All figures trace to params.md v2.0, March 2026.